# Leading Innovation in Complexity

## *Why Strategy, Structure and Culture Must Evolve Together*

A strategic perspective for business and innovation leaders

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## Executive Perspective

Organisations have become increasingly sophisticated in managing innovation. They develop strategies, governance structures and portfolio processes, yet initiatives still encounter unexpected obstacles, conflicting priorities and resistance during implementation. These difficulties are often attributed to complexity, without examining what actually makes innovation complex.

Innovation brings together technological, commercial, organisational and human factors whose interactions are difficult to predict. A decision that appears sensible within one function may create problems elsewhere. Stakeholders interpret the same situation through different assumptions, while established cultures influence which solutions are considered acceptable.

The challenge is therefore not merely to design better processes. It is to understand how innovation's different dimensions interact, and how people behave within them.

> **Innovation cannot be managed through strategy, structures and processes alone. Success also depends on how leaders recognise complexity, reconcile different perspectives and address the cultural assumptions shaping organisational behaviour.**

Rather than seeking to eliminate uncertainty through ever more detailed controls, leaders must learn to navigate it while maintaining strategic direction.

## 1. Why Innovation Becomes Complex

Complexity is not simply another word for difficulty. It arises when multiple variables interact, outcomes remain uncertain and interventions have consequences that cannot be reliably predicted.

Consider the transition to a more sustainable energy system. Technology influences infrastructure investment; infrastructure affects economic incentives; incentives influence consumer behaviour; and political decisions reshape the conditions for investment. Changes in one area can produce unexpected consequences elsewhere.

Organisational innovation displays similar interdependencies. Five factors are especially relevant.

### Five sources of complexity

**Cross-functional interdependence**. Innovation involves engineering, operations, commercial functions, finance and often external partners. Each has legitimate but different priorities.

**Interconnected building blocks**. New propositions combine technological, operational and commercial components whose value depends on how well they work together.

**Development sequence**. The order of design, testing and implementation matters. A technically successful solution may fail when supporting processes or customer readiness lag behind.

**Uncertainty and unpredictability**. Assumptions about technology, markets, costs and organisational readiness can be tested, but some consequences emerge only during implementation.

**Organisational culture**. Professional groups and business units develop different beliefs about quality, risk, authority and appropriate behaviour.

These factors reinforce one another. A technological change may require new processes, which alter responsibilities and challenge established assumptions about expertise. What appeared to be a technical initiative becomes an organisational challenge.

### Why planning alone is insufficient

Planning, accountability and control remain essential, particularly where investments and operational risks are substantial. But they cannot eliminate uncertainty when relationships between variables are not fully understood.

A project may achieve its technical milestones yet fail commercially or organisationally. The cause may not be poor project execution; the plan may have assumed more predictability than the situation allowed.

> **The objective of innovation leadership is not to eliminate complexity, but to create conditions in which it can be understood and navigated.**

## 2. The Four Dimensions of Innovation

Innovation is often managed through projects, budgets and milestones. Yet project outcomes depend on decisions made at several interconnected organisational levels.

The Innovation Dimension Framework (Klokgieters, 2014) identifies four dimensions: Context, Innovation Strategy, Innovation Operating Model and Innovation Execution.

### Context: understanding the changing environment

Technological, economic, societal, regulatory and competitive developments create opportunities and threats. Understanding context means exploring how these developments interact and what alternative futures imply for customers, capabilities and competitive advantage.

### Innovation Strategy: making deliberate choices

Innovation strategy translates this perspective into choices about markets, propositions, business models and capabilities. It determines the balance between strengthening the existing business and exploring future opportunities. The important issue is not a universal investment ratio, but whether the portfolio reflects conscious choices about value, uncertainty and ambition.

### Innovation Operating Model: organising for innovation

The operating model aligns resources, responsibilities, capabilities and decision-making with strategic ambition. An organisation may aspire to radical innovation while retaining incentives and governance designed almost exclusively for operational efficiency. Its own design then becomes a constraint.

### Innovation Execution: turning choices into outcomes

Execution encompasses experimentation, development, implementation and learning across the innovation portfolio. Delays or resource conflicts may appear to be project problems, while their causes lie in strategy or operating-model choices.

### Alignment is more important than sequence

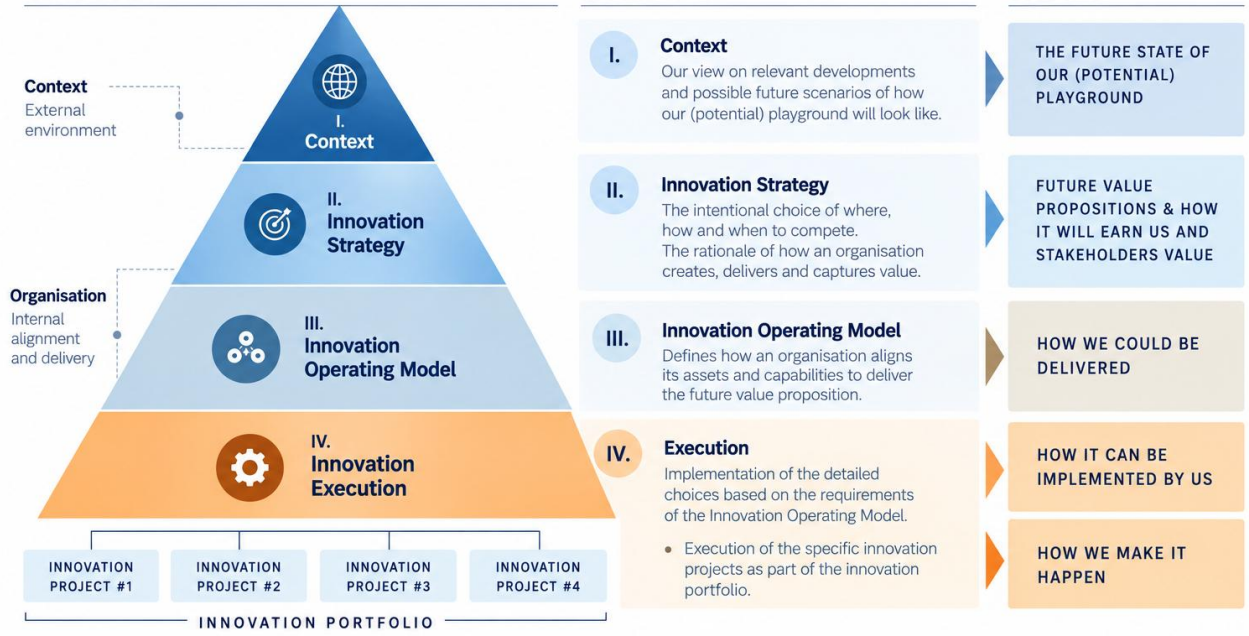
The four dimensions are interconnected, not a one-way cascade. Execution generates learning that can challenge strategic assumptions; a new strategy may require different organisational capabilities and leadership behaviours.

FIGURE 1

## Innovation Dimension Framework

From context to execution – creating value through aligned innovation decisions

The Innovation Dimension Framework shows how context, strategy, operating model and execution are connected and together determine the future value of innovation.




The framework offers senior management a diagnostic question: when innovation struggles, are the four dimensions aligned, or are project teams being asked to compensate for contradictions elsewhere in the organisation?

## 3. Different Perspectives, Different Decisions

Two leaders can examine the same market development and reach opposite conclusions. One sees a competitor to defend against; another sees a potential collaborator. The information may be identical, but their assumptions differ.

Perspectives arise from professional experience, responsibilities, values and beliefs. An operations manager may prioritise standardisation, while an innovation leader emphasises experimentation. A finance director may seek predictable returns, while an engineer sees the need to explore before outcomes can be quantified.

These are not necessarily incompatible views. Problems emerge when each party treats its interpretation as objective reality.

### What lies behind the disagreement?

Imagine a management team debating whether to develop a technology internally or work with a partner. The apparent question is make-or-buy. Underneath may lie conflicting assumptions about control, trust, intellectual property and what creates competitive advantage.

More data will not necessarily resolve a disagreement rooted in unexamined assumptions. Leaders need to make those assumptions visible before choosing a solution.

Diversity of perspectives is a potential source of innovation because it challenges established thinking. Yet without constructive dialogue it can also produce conflict, fragmentation and delayed decisions.

> **The ability to recognise, challenge and combine different perspectives is itself an innovation capability.**

## 4. The Hidden Influence of Corporate Culture

Organisations often respond to innovation challenges by redesigning structures, clarifying responsibilities or introducing new management systems. These interventions are visible and relatively straightforward to document. Their effectiveness, however, depends on less visible cultural factors.

Edgar Schein distinguishes three levels of organisational culture: visible artefacts and behaviour, espoused values and beliefs, and underlying basic assumptions. The original article adds a fourth, distinct layer: formal organisational structure.

### Four layers that shape behaviour

Formal organisational structure defines responsibilities, decision rights, reporting relationships, procedures and management systems.

Visible symbols and behaviour include language, rituals, communication patterns and everyday interactions.

Values and beliefs express what people consider desirable, legitimate and important.

Basic assumptions are deeply embedded beliefs that employees often take for granted. They are difficult to recognise precisely because they rarely need to be articulated.

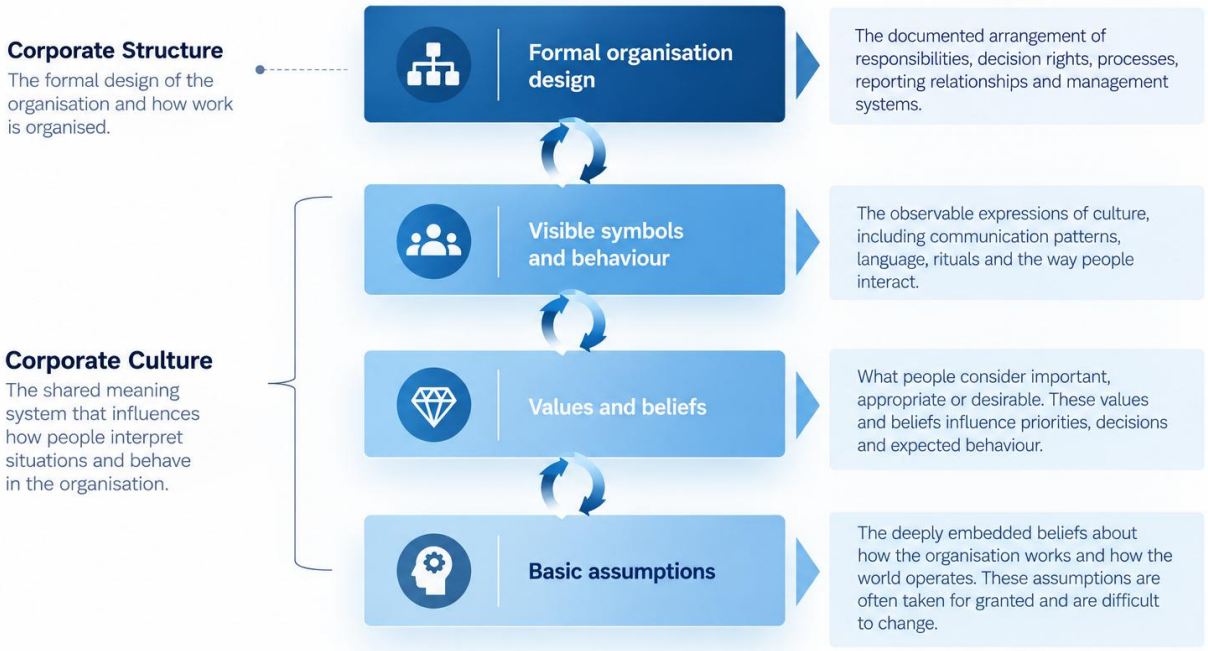
A company may formally encourage experimentation while employees assume mistakes damage their reputation. It may introduce cross-functional teams while managers continue to believe their primary obligation is to protect departmental results.

FIGURE 2

## Corporate Structure versus Corporate Culture

Formal structures and underlying cultural layers influence each other

The formal organisational structure is visible and can be changed directly. Organisational culture operates through visible behaviours, values and underlying assumptions, which are less visible but strongly influence how people act.




Structure and culture influence one another. Formal arrangements can reinforce or challenge cultural patterns, while cultural assumptions determine how those arrangements are interpreted. Organisational redesign therefore does not automatically produce behavioural change.

### The paradox of strong cultures

A strong culture can provide identity, consistency and efficient coordination. Yet the assumptions supporting today's performance may also limit tomorrow's innovation. Established beliefs shape which ideas are credible, which risks are acceptable and whose expertise is trusted.

The leadership challenge is not to weaken culture indiscriminately. It is to distinguish the assumptions worth preserving from those that prevent adaptation.

> **The stronger the culture supporting today's success, the harder it may be to challenge the assumptions on which that success rests.**

## 5. Leading Innovation Across Boundaries

A personal experience early in my career illustrates the practical consequences of misalignment between structure and culture.

### The APT experience: one system, different assumptions

In 1986, I joined AT&T-Philips Telecommunications (APT), following the combination of telecommunications activities from AT&T and Philips. One of my responsibilities was introducing a common project planning tool in the R&D organisation.

The objective appeared straightforward: integrate technologies, products and working methods through a standardised planning system. Structurally, the proposal made sense. In practice, the teams brought very different assumptions about good product development.

The former Philips teams emphasised technological quality, craftsmanship and professional expertise. Authority was closely associated with technical knowledge and seniority. The former AT&T teams placed greater emphasis on speed to market, commercial performance and business outcomes.

These differences influenced how teams understood priorities, project plans and management responsibilities. A common tool could standardise procedures, but it could not by itself reconcile different beliefs about quality, speed, authority and success.

The implementation encountered resistance because the design had focused on formal arrangements without sufficiently addressing the underlying professional cultures.

The lesson was not that standardisation was undesirable. It was that a common structure does not automatically create a common understanding of how people should work.

### From project control to organisational coherence

Innovation leaders often encounter similar situations. New processes are introduced to improve collaboration, new governance arrangements to accelerate decisions and new technologies to integrate activities. Such interventions may be necessary, but they succeed only when the perspectives and behaviours shaping adoption are understood.

Leading innovation therefore requires coherence between strategic ambition, organisational capabilities, formal structures, cultural assumptions and execution. Where these reinforce one another, innovation can progress. Where they conflict, even well-designed initiatives may struggle.

## Leadership Reflection

1. **Understanding complexity**. When innovation struggles, do we investigate interdependencies and uncertainty, or focus mainly on project performance?

2. **Alignment**. Are Context, Innovation Strategy, Innovation Operating Model and Innovation Execution mutually supportive?

3. **Perspectives**. Which assumptions shape our management team's interpretation of innovation opportunities, and which might limit our choices?

4. **Culture and structure**. Do formal arrangements reinforce the behaviours innovation requires, or do cultural assumptions undermine them?

5. **Leadership**. Are we mainly managing innovation projects, or also managing the organisational conditions that enable them to succeed?

These questions are intended to expose assumptions and stimulate executive dialogue, not to produce instant consensus.

## Closing Perspective

Innovation is complex because technologies, markets, capabilities, structures and people interact in ways that cannot always be predicted. More detailed plans and sophisticated systems remain useful, but cannot resolve every contradiction.

Innovation also demands leaders who recognise different perspectives, understand cultural assumptions and align strategic ambition with organisational reality. The choice is not between discipline and exploration. It is about developing an organisation capable of combining both.

> **The effectiveness of innovation leadership is determined not by how completely complexity is controlled, but by how effectively the organisation learns to navigate it.**

## About the Author

Koen Klokgieters is the founder of InnoPlot and has spent more than three decades at the intersection of strategy, innovation, technology and organisational transformation. His experience combines corporate leadership, consulting, entrepreneurship and education.

Through InnoPlot, he supports organisations and their leaders in translating strategic challenges into innovation capabilities, organisational change and measurable impact.

InnoPlot | Business & Technology Innovation (innoplot.com)

## Conceptual Foundations

Adapted from Innovation, Complexity, Dimensions, and Perspectives by Koen Klokgieters. The Innovation Dimension Framework is attributed to Klokgieters (2014). Corporate Structure versus Corporate Culture builds on the author's extension of Edgar Schein's organisational culture model.

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Schein, E.H. (1985). Organizational Culture and Leadership. Hoboken, NJ: Wiley.